

Demonstrations Involving Photography

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SINCE every third freshman is probably an enthusiastic amateur photographer, demonstrations involving chemistry in photography should be welcomed. A number of such experiments will be described; with slight modifications they may be performed by the students themselves.

The lecture room need not be light-tight; pulling down the shades and substituting amber for white lights is sufficient.

1. DEMONSTRATIONS EXPLAINING THE CHEMISTRY OF PHOTOGRAPHY

(A) *Development* (Illustrating an active reducing agent converting Ag^+ to Ag^0). Bright light. To 100 cc. $N/100 \text{ AgNO}_3$ add 100 cc. $N/100 \text{ NaCl}$. Before the suspension of AgCl has a chance to separate out, add 10 cc. inorganic developer (a) (See formulas at end of article). Reduction to metallic silver should take place almost instantaneously at 20°C . An inorganic developer is chosen here because of simplicity of explanation.

(B) *Fixing* (Illustrating the solvent effect of $\text{Na}_2\text{S}_2\text{O}_3$ on AgCl). Bright light. To 100 cc. $N/100 \text{ AgNO}_3$ add 100 cc. $N/100 \text{ NaCl}$. Then quickly add 60 cc. $N/3 \text{ Na}_2\text{S}_2\text{O}_3$. At 20°C ., clearing should take place in about three seconds. A discussion as to the chemical and physical nature of this solution is advantageous.

(C) *Normal Printing Processes*. In advance prepare a synthetic negative as follows: On one half of a piece of glass which will fit a printing frame ($4'' \times 5''$ or larger, depending on size of audience) paste gummed black letters to read "Black on white," while on the other half, with black paint, blacken out all except the letters "White on black." Amber light. Expose a sheet of *Azo F21* paper behind the negative as follows: Distance from 100-watt white light bulb without reflector 6 feet. Time, 5 seconds. The gummed letters should be in contact with the sensitive side of the photographic paper. Develop in metol developer (b), or any standard paper developer (D64, D72, etc.) for 3 minutes at 20°C . Rinse in acid short stop bath (roughly $N/5$ acetic acid). Then "fix" in fixing bath (c) or any standard fixing bath, for 3 to 5 minutes, after which, if all other unexposed sensitive paper has been covered, the bright light can be turned on. The result will be a paper negative of the glass negative, making a positive, as the legend will show.

This experiment can be modified as follows:

1. The glass negative can be printed when it is reversed (*i. e.*, when the gummed letters are not in contact with the sensitive side of the paper) which will cause the legend to read from right to left.

2. Proper exposure time and/or distance can be

determined by varying one at a time and developing for the proper time.

3. Fogging effect caused by leaving the properly exposed print in the developer too long can be shown. This will, of course, illustrate that even the unexposed silver halide is reducible by the developer.

(D) *Reversal Printing* (Illustrating the making of a positive directly, a method used in processing amateur cinematographic film and Dufay color films). Repeat C exactly as above, up to the immersion in the short stop bath. Remove the developed print from the developer, wash in water, then place in bleaching bath (d), leaving it there until all of the black Ag^0 has been oxidized to soluble Ag_2SO_4 . Remove, rinse in water, and immerse in a 2–5 per cent NaHSO_3 solution to remove any MnO_2 stain formed. Now the bright lights may be turned on. Expose the bleached print, which still has light-sensitive silver halides on it, to bright light (in this case exposure time, distance, etc., are not critical), develop (here developing time is not of great consequence), rinse, fix, and wash. This should lead to a print which is truly "Black on white" and not a negative as was the case in C.

(E) *Toning* (Illustrating the replacement of metallic Ag by other metals or by colored insoluble salts). Preliminary experiments may consist in demonstrating that (1) Ag^0 plates out Au^0 from a borax solution of AuCl_3 ; (2) $\text{Fe}^{+3} + \text{Fe}(\text{CN})_6^{-4} \rightarrow$ blue ppt.; (3) $\text{Cu}^{+2} + \text{Fe}(\text{CN})_6^{-4} \rightarrow$ red ppt.; (4) $2\text{Ag}^0 + \text{S}^{-2} \rightarrow \text{Ag}_2\text{S}$, brown-black. Then show the application of these facts by the following photographic processes.

Expose, develop, fix, and wash a sheet of paper exactly as indicated in C except that the glass negative is omitted. Cut into many sections and in bright light perform these toning operations:

1. *Gold Toning*.—Place numbered or labeled sections of the black print into gold-toning bath (e) for 1, 2, 3 ... 10 minutes at 20°C . Wash thoroughly and dry; and note how the color of the deposit has changed from black through red-brown to purplish black.

2. *Iron (Blue) Toning*.—Place numbered sections in iron-toning solution (f) for 1, 3, 5, 7, and 9 minutes at 20°C . Remove, rinse, and note how the dried prints range from black to blue. This illustrates that $\text{K}_3\text{Fe}(\text{CN})_6$ oxidizes Ag^0 to insoluble colorless $\text{Ag}_3\text{Fe}(\text{CN})_6$, which, by double decomposition, reacts with Fe^{+3} to form insoluble $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$.

3. *Copper (Red) Toning*.—Place numbered sections in copper-toning solution (g) for 1, 3, 5, 7, and 9 minutes at 20°C . Wash in distilled water. The tones should range from cold brown to red. As before, the Ag^0 is oxidized, and by reaction with Cu^{+2} , forms insoluble red cupric ferrocyanide.

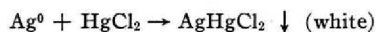
4. *Sulfide (Sepia) Toning*.—Place numbered sections in 20 per cent solution of $\text{Na}_2\text{S}_2\text{O}_3$ for 5 minutes at 20°C .,

¹ From here on, all exposure and development times will be mentioned in terms of this paper. Correct exposure for other papers can be determined readily.

then, *without* rinsing, immerse in 3 *N* HCl for 15, 30, 45, and 60 minutes at room temperature. The finely divided S formed on acidification of the $\text{Na}_2\text{S}_2\text{O}_3$ oxidizes the metallic Ag^0 to insoluble Ag_2S , sepia.

(F) *Intensifying the Silver Deposit.* Amber light. The film needed is 35-mm. "positive" film, which is slow and non-panchromatic. By means of resistance cut down the intensity of a white electric light bulb until the proper exposure, at say 10 feet, is 32 seconds, when developed in developer (b) or any other film developer (DK20, D76, etc.) for 3 minutes at 20°C. The determination of proper time and distance of exposure is best done by the "hunt and pick" method. Having found the proper conditions, cut off seven inches of positive film and expose the entire strip for one second, move an opaque card down one inch and expose the remaining six inches for another second; move the cover down another inch and expose the remaining five inches for two seconds more, move down an inch and expose for four additional seconds, another inch and expose for eight seconds more, another inch for an additional 16 seconds, and finally expose the last inch for an additional 32 seconds. The film then has been exposed in a systematic method from one to 64 seconds as follows: 1, 2, 4, 8, 16, 32, and 64 seconds. Also expose another strip of positive film under the same conditions, but only for 8 seconds. This latter strip will be decidedly underexposed. Develop both strips simultaneously in the same developer for 3 minutes at 20°C. Rinse, fix, and wash. One strip should be uniformly gray, because it is underexposed, while the second should be increasingly darker in graduated steps, where the exposure of any given step is related to that of its neighbors by a factor of 2. Now bright lights can be turned on. Cut the uniform gray negative into four parts.

1. *Mercury Intensification.*—Thoroughly wet one section of the gray negative and immerse in the mercury bleach (h) until clear. (Care! This solution contains HgCl_2 , very poisonous, and also very irritant to sensitive skins. Perhaps use rubber gloves.) The reaction is:



Then wash the film and place in developer (b) for 3 minutes at 20°C. Here the organic reducing agent in the developer reduces the silver mercurous chloride to metallic mercury (and perhaps silver too). Rinse, fix, and wash; and compare the dry negative with the original. Even though only one atom of mercury were formed for an atom of silver, the negative would be more dense, for black mercury is more opaque than black silver. Casual visual inspection may not show that the intensified negative is actually more opaque—it may take the third part of this demonstration to show that such is the case.

2. *Physical or Silver Intensification.*—The bath used here has AgNO_3 and a mild organic reducing agent, which thus liberates colloidal Ag^0 slowly. These particles form around the silver deposit on a negative more

easily than they do in the solution or on the clear places on the negative. Thus if a negative is immersed in this bath, the deposit of silver on the negative increases, apparently without a change in contrast.

Place two more developed and fixed sections of the thin negative in the *freshly made* physical intensifying bath (i) leaving one for 10 minutes and the other for 20 minutes at 20°C. Rinse well, then immerse in a 30 per cent solution of $\text{Na}_2\text{S}_2\text{O}_3$ for two minutes. Wash thoroughly and dry. Too long immersion of a developed film in "hypo" might decrease the amount of silver deposit.

The two physically intensified negatives can be compared with the original control to see whether there has been an increase in the opacity. Perhaps conclusions can be drawn as to the relative merits of the two intensifying processes. Label each of the four sections and save for part 3 below.

3. *Proof of Intensification.*—The proof should be in the printing. In amber light, place as negatives the step wedge, the control, and the three intensified sections in intimate contact with a sheet of sensitive paper. This can be done between two pieces of clean glass, or in a regular printing frame. Properly expose to white light, develop, and fix. Now by comparison with the step wedge and with each other, conclusions can be drawn as to whether intensification actually took place, and at least roughly quantitatively, as to the merits of the two processes. Remember that going from one step in the wedge to the next darker means doubling the deposit of silver. The step wedge is saved for future experiments.

(G) *Reducing the Quantity of Silver Deposit.* It frequently happens that, due to incorrect exposure or incorrect development, or both, the negative is too dense. The contrast of the negative may or may not be correct, but the silver is too thick, thus making for an excessively long printing time. Not only can the amount of silver per square centimeter be lessened, but it can be done proportionally (no change in contrast), superproportionally (decreasing contrast), or subproportionally (increasing contrast).

In amber light, expose four strips of positive film exactly as in F above except that if it were found that 32 seconds was the correct exposure time, expose half of each film for 64 seconds and the other half for 16 seconds. After development in a developer for three minutes at 20°C., rinse, fix, and wash. Part of each negative will be overexposed, the other part underexposed, the contrast being 1:4. Now intensification may be performed in bright light.

1. *Subproportional Reduction.*—Place one of the negatives in subproportional reducer (j) for 1–4 minutes at 20°C., or until noticeable reduction has taken place. Wash and dry. The $\text{K}_3\text{Fe}(\text{CN})_6$ has again formed $\text{Ag}_3\text{Fe}(\text{CN})_6$ which is soluble in the NH_4CNS , also a constituent of the reducer. Thus some silver dissolves.

2. *Superproportional Reduction.*—Place the second of the negatives in superproportional reducer (k) for 1–4

minutes at 20°C., then wash and dry. The $(\text{NH}_4)_2\text{S}_2\text{O}_8$ oxidizes the Ag^0 to soluble Ag_2SO_4 .

3. Proportional Reduction.—Place the third of the negatives in proportional reducer (*l*) for 1–3 minutes, then remove any MnO_2 stain by immersion for a minute or so in a one per cent solution of sodium metabisulfite. Rinse well and dry. This reducer, as the formula shows, is really a mixture of two reducers, one for increasing contrast and the other decreasing contrast.

4. Proof of Reduction and Contrast Change.—In amber light place the three reduced negatives, the control, and the step wedge in the printing frame, in contact with a sheet of photographic paper. Properly expose to white light, develop in paper developer for 3 minutes at 20°C., and compare the results. Not only should the prints of the reduced negatives be darker, but as ascertained by the step wedge, there should be a change in contrast in two cases.

(H) *Commercial Silver Recovery from "Exhausted" Fixing Bath.* (Illustrating the electrochemical series and electrodeposition.) Bright light. To some greatly used fixing bath (1) add 1–2 g. Na_2S per liter of bath, forming extremely insoluble Ag_2S , (2) add metallic zinc, (3) electrodeposit the silver on a cathode. Negative film contains about one mg. and positive film about 0.3 mg. of Ag per cm^2 , 75 per cent of which finds its way eventually into the fixing bath.

2. CHEMICAL PRINCIPLES EXPLAINED BY PHOTOGRAPHIC EXPERIMENTS

(A) Rates of Reaction.

1. Effect of Temperature, with Glycine Developer.—Amber light. Expose a sheet of photographic paper for 5 seconds, 5 feet from a 100-watt bulb. Cut into five strips and label on the back with a lead (not indelible) pencil as follows: 10°C., 15°C., 20°C., 25°C., and 35°C. Develop in 300 cc. of glycine developer (*m*) to which one per cent KBr has been added to the extent of one part bromide solution to 15 parts water, for precisely 3 minutes at exactly the temperatures indicated above. The same developing solution can be used merely by heating or cooling, as needed. Rinse the developed paper in short stop ($N/5$ acetic acid). Fix and wash all sections in *exactly* the same way. When dried there should be a noticeable difference in the amount of silver produced.

2. Effect of Temperature, on a Number of Developers.—Repeat 2A-1, using metol developer (*b*). Everything else should be exactly the same. When the prints are dried, find one in the glycine series which is just as black as one in the metol series. Call these your standards. Then see whether the metol and glycine prints, developed in each case at 5°C. lower temperature than the standard, are also equally black. If not, the "temperature coefficients" of the two developers are not the same, at least in that temperature range.

3. Effect of Concentration.—In these demonstrations the only variable will be the concentration of the developer. In amber light, properly expose a sheet of photographic paper. Develop labeled sections for 3

minutes at 20°C. in glycine developer (*m*) to which 10 per cent KBr solution has been added to the extent of one part solution to 15 parts developer, and which has been subsequently diluted successively so that the solution employed is $1/2$, $1/4$, $1/8$, $1/16$, and $1/32$ of its original strength. Rinse, fix, and wash as usual. A marked change should be noted in the amount of black silver produced.

If this experiment were repeated with the glycine developer (*m*) without the bromide, interesting results might be obtained. Not only would the most dilute developer produce the least amount of silver, but the color of the silver might be sepia, in place of black. That is probably due to the size of the silver particles produced, colloidal silver having many different colors.

4. Effect of pH. Positive Catalysis.—This is really a demonstration to show the effect of a pH change on development rate. Amber light. Properly expose a sheet of paper. Cut into sections, which are then numbered. Develop the first in 250 cc. of a non-buffered developer (*n*) for 3 minutes at 20°C. Rinse, fix, and wash. Develop the second section in the same developer, to which has been added 2 cc. concentrated HCl, maintaining all other factors constant. Rinse, fix, and wash. Repeat using 2 cc. more of the concentrated HCl, and finally 4 cc. more HCl. Even when 10 cc. of concentrated HCl have been added the pH is still greater than 7. For those interested in photography, this demonstration shows the need for controlling the alkalinity of the developer, and explains why immersion of the print, wet with developer, into the acid short stop, stops development. For all of the students, it shows that this particular reducing agent works faster, the more alkaline the solution.

5. Effect of KBr. Inhibition.—Those who develop their own films know that the presence of KBr in the developer appreciably slows down development rate. That fact will be used here to illustrate negative catalysis, although that is probably only part of the answer.

Amber light. Expose a sheet of paper properly. Cut into sections and label. Develop one strip for 3 minutes in 250 cc. half-strength developer (*m*) to which has been added $1/4$ cc. of 10 per cent KBr. Rinse, fix, and wash. Develop the other strips exactly the same way, except that KBr solution has been added in $1/4$ -cc. steps until the final developer has in it 1.5 cc. KBr solution.

(B) Quantity of Silver Produced.

1. Effect of Exposure Time.—In amber light, properly expose a small piece of the same paper, at the same distance from the same bulb, but at $1/10$, $2/10$, $3/10$. . . $9/10$ the proper exposure. Develop all of them in the same paper developer for exactly the same length of time and at the same temperature. Rinse, fix, and wash. The photographers in the class will recognize what they already know, *i. e.*, that the amount of Ag^0 produced depends on exposure time, all other factors being kept constant. Those not particularly interested in photography will see that the intensity of the latent image

produced is a function of the energy received from the light source.

2. Effect of Color of Light. Photosensitizers.—In absolute darkness, expose a sheet of paper to the light filtered by a deep amber filter (cellophane or glass) for a few seconds. In complete darkness develop properly in a paper developer, rinse, fix, and wash, turning on the amber light as soon as the paper has been in the fixing bath for three minutes.

Repeat exactly, but before exposing the paper to the filtered light, sensitize it. This is done by swabbing the paper for several minutes in absolute darkness with sensitizing solution (o), after which wash well with distilled water, then immerse in 95 per cent alcohol and dry rapidly (all operations in absolute darkness, of course).

Those students who are familiar with photographic principles realize that this experiment demonstrates the making of panchromatic and orthochromatic films from

emulsions blind to certain radiations. The theory of sensitizers and desensitizers is given in Clerc, Neblette, and Henney-Dudley, as well as in other standard texts on photography.

3. Effect of Ammonia.—Expose a sheet of paper for one-tenth the proper exposure. Cut into three sections. Paint three desiccators with black asphaltum until they are light-tight. Place in one desiccator concentrated H_2SO_4 , in the second a few cc. of concentrated NH_4OH , and in the third water. Allow a piece of the exposed paper to remain in each desiccator for a short time. Then develop the three sections in exactly the same way, in the same developer for the same length of time. Rinse, fix, and wash by the same procedure. There should be a change in the quantity of silver produced. The use of NH_3 in increasing the speed of an emulsion is used commercially with films, although the NH_3 is usually applied before, rather than after the exposure.

FORMULAS

(a) Inorganic Developer

Sodium bisulfite.....	30.0 g.
Sodium hydrosulfite.....	20.0 g.
H_2O to make.....	1.0 liter

(b) Metol Developer

H_2O (52°C.).....	500.0 cc.
Metol.....	6.2 g.
Sodium sulfite, anhydrous.....	45.0 g.
Sodium carbonate, anhydrous.....	67.5 g.
Potassium bromide.....	1.9 g.
H_2O to make.....	3.0 liters

(c) Fixing Bath

Water.....	2.0 liters
Sodium thiosulfate.....	480.0 g.

When thoroughly dissolved, add the following solution slowly to the cool thiosulfate solution while stirring the latter rapidly:

Water at 52°C.....	160.0 g.
Sodium sulfite, desiccated.....	30.0 g.
Acetic acid (28% pure).....	96.0 cc.
Potassium alum.....	30.0 g.

(d) Bleaching Bath

An aqueous solution which contains 0.4% of both KMnO_4 and H_2SO_4 .

(e) Gold-Toning Bath

Sodium acetate, fused.....	10.0 g.
Borax.....	1.0 g.
Gold chloride (1% solution).....	25.0 cc.
H_2O to make.....	1.0 liter

(f) Iron-Toning Bath

Ferric ammonium citrate (10% solution).....	60.0 cc.
$\text{K}_3\text{Fe}(\text{CN})_6$ (10% solution).....	60.0 cc.
Acetic acid (10% solution).....	600.0 cc.

(g) Copper-Toning Bath. Mix equal volumes of No. 1 and No. 2

No. 1	No. 2
$\text{K}_3\text{Fe}(\text{CN})_6$	CuSO_4
Potassium citrate... 3.2 g.	Potassium citrate... 4.0 g.
Potassium citrate... 16.0 g.	Potassium citrate... 16.0 g.
H_2O to make..... 600 cc.	H_2O to make..... 600.0 cc.

(h) Mercury Bleach

Potassium bromide.....	22.5 g.
Mercuric chloride.....	22.5 g.
H_2O to make.....	1.0 liter

(i) Physical Intensifying Bath

Crabtree and Muehler² recommend:

Solution 1

Silver nitrate.....	60.0 g.
H_2O to make.....	1.0 liter

Solution 2

Sodium sulfite (desiccated).....	60.0 g.
H_2O to make.....	1.0 liter

Solution 3

Sodium thiosulfate crystals.....	105.0 g.
H_2O to make.....	1.0 liter

Solution 4

Sodium sulfite (desiccated).....	15.0 g.
Metol.....	24.0 g.
H_2O to make.....	3.0 liters

To prepare the intensifier, add one part of solution (1) slowly to one part of solution (2), obtaining a thorough mixture by stirring or shaking. A white precipitate will form which is dissolved by the addition of one part of solution (3). The resulting solution should be permitted to stand until clear, after which three parts of solution (4) should be added, with constant stirring. The film should be treated immediately in this solution. This mixed solution is stable for about 30 to 45 minutes at 20°C.

(j) Subproportional Reducer

Potassium ferricyanide.....	5.0 g.
Ammonium sulfocyanate.....	10.0 g.
H_2O to make.....	1.0 liter

(k) Superproportional Reducer

H_2O	500.0 cc.
Ammonium persulfate.....	60.0 g.
Concentrated H_2SO_4	3.0 cc.
H_2O to make.....	1.0 liter

(l) Proportional Reducer

NIETZ AND HUSE REDUCER

Solution 1

Potassium permanganate.....	0.25 g.
Sulfuric acid (10%).....	15.0 cc.
Distilled water to make.....	1.0 liter

² CRABTREE AND MUEHLER, "Reducing and intensifying solutions for motion picture film," *J. Soc. Motion Picture Engrs.*, 17 (No. 6), 1001-53 (1931).

Solution 2

Ammonium persulfate.....	25.0 g.
Distilled water to make.....	1.0 liter

For use, one part of solution (1) is added to three parts of solution (2). The keeping properties of the combined solutions are very poor so the solutions should be mixed immediately before use.

(m) Glycine Developer	
Sodium sulfite, anhydrous.....	15.0 g.
Potassium carbonate.....	45.0 g.
Glycine.....	10.0 g.
H ₂ O to make.....	1.0 liter
At time of use dilute to 1/6 strength	

(n) Non-buffered Developer	
Sodium sulfite, anhydrous.....	187.0 g.
Metol.....	21.5 g.
Hydroquinone.....	43.0 g.
Sodium hydroxide.....	43.0 g.
H ₂ O to make.....	1.0 liter

(o) Sensitizing Solution	
Ethyl alcohol (95%).....	40.0 cc.
Pinacyanol stock solution (0.1%).....	4.0 cc.

Concentrated NH ₄ OH.....	2.0 cc.
H ₂ O.....	60.0 cc.

Editor's Note: Readers may be interested in referring to previous articles on photography which have appeared in the JOURNAL OF CHEMICAL EDUCATION:

- ALYEA, "Lantern slide technics," **16**, 308 (1939).
 CARROLL, "The preparation of photographic emulsions," **8**, 2341 (1931).
 DUNBAR, "The photographic club," **14**, 91 (1937).
 FLIEDNER, "A chemistry club project in photography," **11**, 620 (1934).
 GOLDBLATT AND BATES, "Preparation of lantern slides without a camera," **17**, 462 (1940).
 GURINSKY, "A camera for photographic demonstration," **17**, 432 (1940).
 MACLEAN, "A project for general chemistry students: color toning of photographic prints," **14**, 31 (1937).
 MURRAY AND HANSON, "Coloring snapshots with organic dyes," **17**, 430 (1940).
 SCHUMACHER, "Photography in the college curriculum," **17**, 427 (1940).
 SHEPPARD, "The chemistry of photography. I. Historical considerations. II. The latent image. III. Development of negative and positive images," **4**, 298, 465, 749 (1927).
 "The chemistry of photographic materials," **6**, 2002 (1929).